Airline Safety Analysis

Should Travellers Avoid Airlines With Past Crashes?

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1. Dataset Overview

The report uses the FiveThirtyEight Airline Safety dataset which includes data from 56 major commercial airlines for two time periods from 1985 to 1999 and 2000 to 2014. The study examines four variables which include the number of incidents, the number of fatal accidents, the total number of fatalities, and the weekly seat-kilometre availability which is measured in Available Seat-Kilometres (ASK). The study included all 56 airlines in the analysis because there were no missing data points.

Variable	Old Period (1985–99)	New Period (2000–14)
Safety Incidents	inc_old	inc_new
Fatal Accidents	fa_old	fa_new
Fatalities	fat_old	fat_new
Airline Size (ASK/week)	ask_wk	ask_wk

Table 1. Dataset variables.

2. Methodology

2.1 Exposure Normalisation

Raw counts favour large airlines. All metrics are divided by total exposure (ASK/week × 52 × 15), giving rates per billion seat-km — the standard aviation regulator metric. Formula: Rate = Count ÷ (ASK/week × 52 × 15 × 10^9).

2.2 Derived Metrics & Analytical Approach

The researchers computed period-over-period changes (d_r_inc, d_r_fa, d_r_fat) together with the improvement metric (impr = old fatality rate − new). The analysis process consists of four steps which include (1) EDA — charts and period comparisons; (2) Pearson correlation; (3) Wilcoxon Signed-Rank and Spearman Rank Correlation tests — chosen because data is right-skewed and non-normal; (4) Composite Risk Score using recent-period data only.

3. Exploratory Data Analysis — Part 1

3.1 Top 10 Airlines by Raw Incident Count

The raw total numbers show Aeroflot Delta/Northwest and American Airlines as the most hazardous airlines. The large size of these airlines results in greater flight operations which lead to more safety incidents because their operational scale leads to increased flight operations - this finding does not serve as proof of operational safety standards. The application of exposure-adjusted rates results in major changes to the existing ranking system.

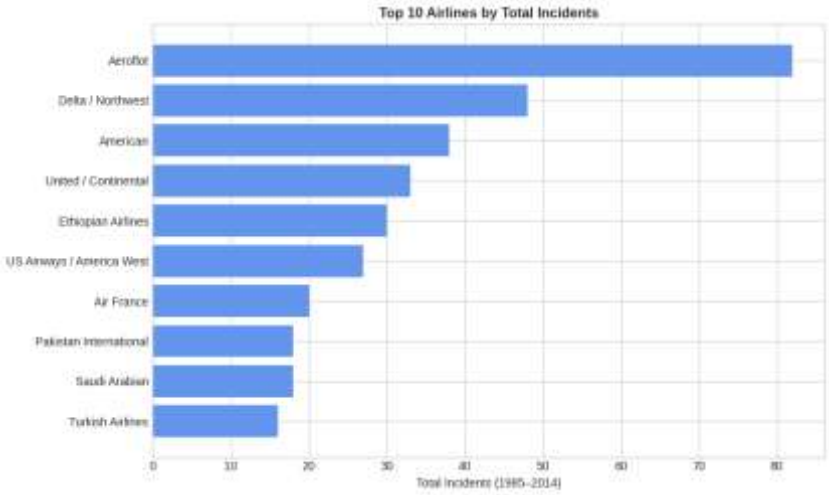


Figure 1. Top 10 airlines by total raw incident count (1985–2014). Size, not danger, drives these rankings.

3.2 Distribution of Incidents — Recent Period (2000–2014)

Incident counts per airline in 2000–2014 are heavily right-skewed. Most airlines had 0–5 incidents while a handful had many more. Mean = 4.12, Median = 3.0. Mean > Median confirms asymmetry, ruling out t-tests in favour of non-parametric alternatives.

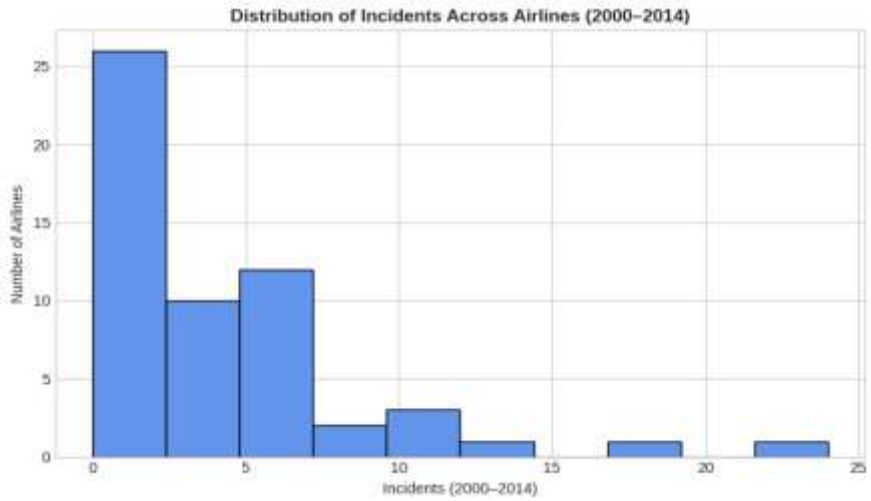


Figure 2. Distribution of incidents per airline (2000–2014). Mean = 4.12, Median = 3.0, confirming right-skew.

3. Exploratory Data Analysis — Part 2

3.3 Period-over-Period Comparison

The safety metrics for all three categories experienced major declines when all 56 airlines were analyzed together as a single entity. The total number of incidents decreased by 42.5% when counting all incidents which dropped from 402 incidents to 231 incidents while fatal incidents decreased by 69.7% from 122 incidents to 37 incidents and fatal injuries dropped by 50.6% from 6295 cases to 3109 cases. Figure 3 presents evidence that demonstrates an overall improvement in safety standards across the entire industrial sector.

Safety Measure	Old Period	New Period	Change
Incidents	402	231	↓ 42.5%
Fatal Accidents	122	37	↓ 69.7%
Fatalities	6,295	3,109	↓ 50.6%

Table 2. Industry-wide safety changes between the two periods.

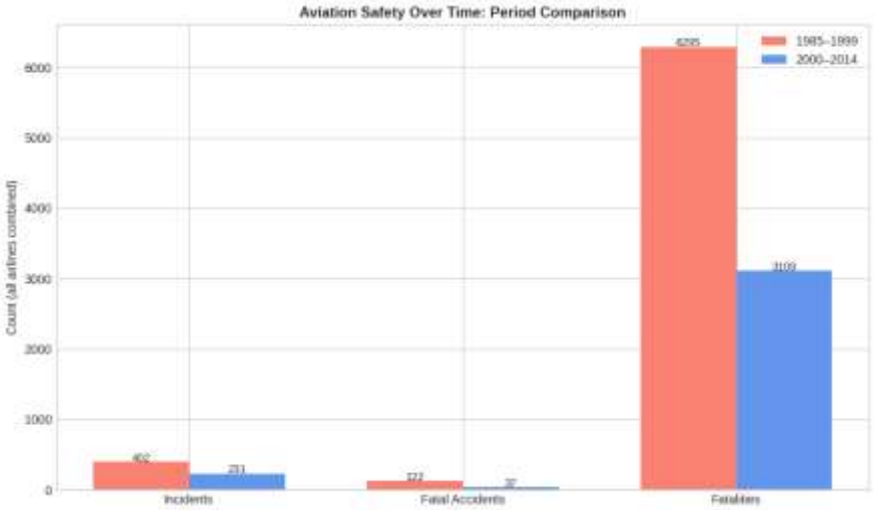


Figure 3. Period comparison across 56 airlines. All three safety metrics declined sharply in the modern period

3.4 Which Airlines Got Safer?

Airlines managed to decrease their deadly incidents which resulted in safer operations. The fatality rates for airlines improved except for three carriers which experienced increased rates, namely Kenya Airways and Malaysia Airlines and Gulf Air. The majority of instances show that Malaysia Airlines experienced two major incidents, MH17 and MH370, which caused their safety record to deteriorate instead of showing ongoing safety issues. The process of identifying a single catastrophic event from ongoing operational breakdowns requires distinct evaluation methods.

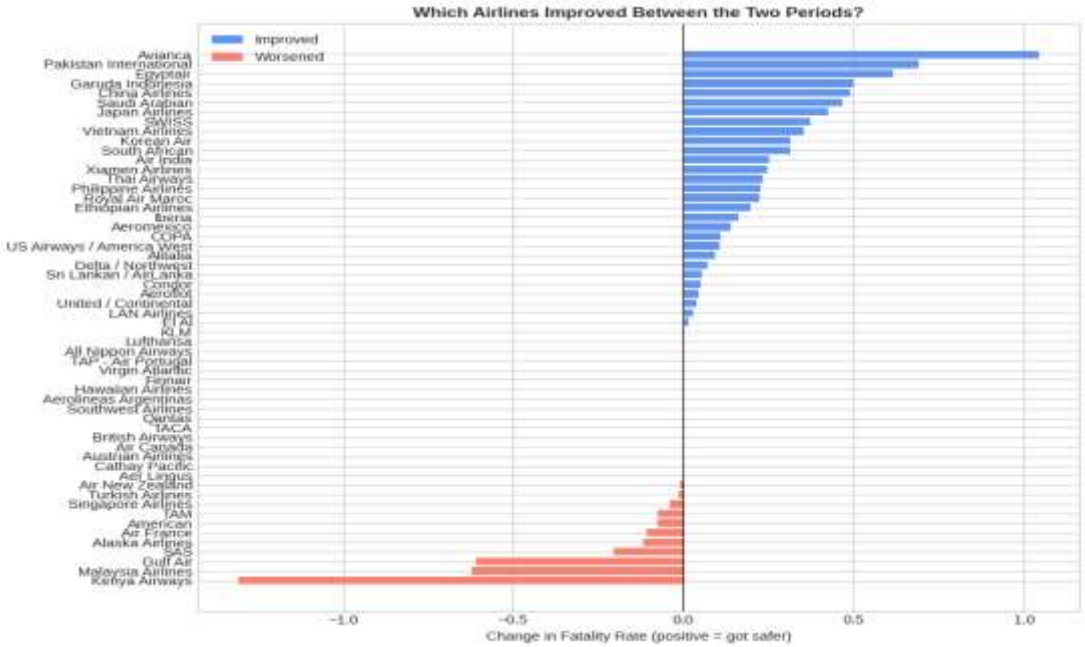


Figure 4. Change in fatality rate per airline between periods. Blue = improved, Red = worsened.

4. Hypothesis Testing — Part 1

Charts suggest patterns but cannot prove statistical significance. Two non-parametric tests were applied because the data are right-skewed and do not follow a normal distribution.

4.1 Pearson Correlation — Past vs Future Safety

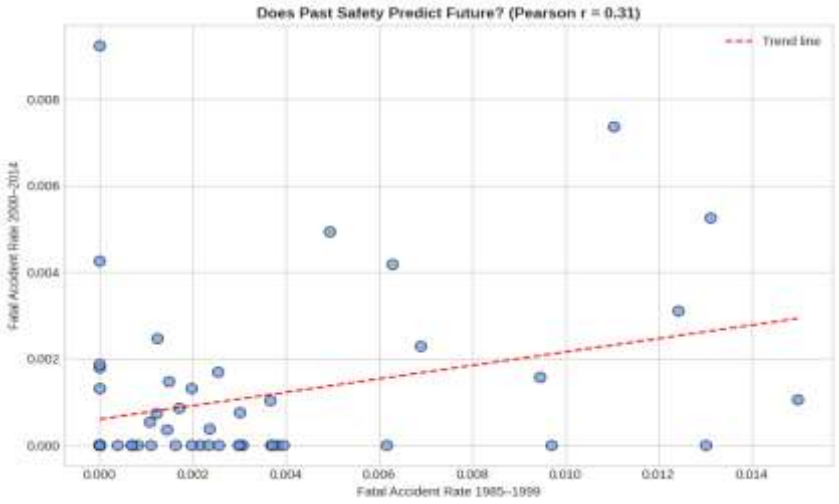


Figure 5. Scatter plot of fatal accident rates (per billion ASK) for 1985–1999 versus 2000–2014 across 56 airlines, showing a weak positive relationship (Pearson r = 0.31) that confirms past safety record is a poor predictor of future performance.

Metric Compared	Pearson r	Interpretation
Raw incidents (old vs new)	0.403	Weak–moderate
Raw fatalities (old vs new)	0.047	Almost none
Incident rate	0.369	Weak
Fatal accident rate	0.311	Weak

Table 3. Pearson correlations between old and new period safety metrics.

4.2 Test 1 — Wilcoxon Signed-Rank Test

The Wilcoxon Signed-Rank test produced results of W = 799.00 and p = 0.0024, which falls below the α threshold of 0.05. The hypothesis should be rejected based on statistical evidence which demonstrates decreased airline incident rates between the two time periods. The histogram, which shows rate differences by airline (Figure 6), indicates that almost all of the rate difference values are negative. The sector shows real safety improvements through its gained results which cannot be attributed to chance fluctuations..

Item	Detail
H_0 (Null)	Incident rates did not change between 1985–99 and 2000–14
H_1 (Alternate)	Incident rates were lower in 2000–14 (one-tailed)
Median rate (old)	0.0057 per billion ASK
Median rate (new)	0.0038 per billion ASK
W-statistic	799.00
p-value	0.0024
Decision	Reject H_0 — significant reduction confirmed

Table 4. Wilcoxon Signed-Rank Test results.

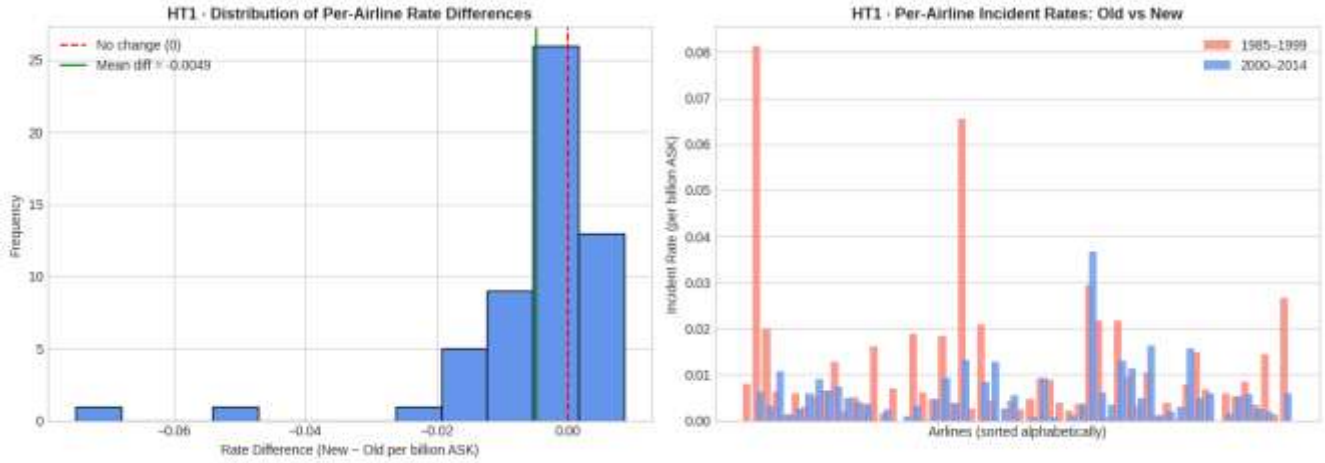


Figure 6. Distribution of per-airline rate differences (new minus old). Most bars negative — widespread improvement. W = 799, p = 0.0024.

4. Hypothesis Testing — Part 2

4.3 Test 2 — Spearman Rank Correlation Test

According to Spearman test on fatal accident rates, $\rho = 0.248$ and $p = 0.0649$. The H0 hypothesis has not been rejected since this is greater than $\alpha = 0.05$. Statistically significantly, there is insufficient evidence to conclude that an airline's past fatal accident rate predicts its future rate. Even though the positive ρ indicates a weak rank-order tendency, it is not statistically significant. The result and the weak Pearson correlations across all metrics generally suggest that the historical safety record is a poor predictor of future performance.

Item	Detail
H_0 (Null)	No monotonic relationship between old and new fatal accident rates ($\rho = 0$)
H_1 (Alternate)	Significant positive rank correlation ($\rho > 0$)
Spearman ρ	0.248
p-value	0.0649
Decision ($\alpha = 0.05$)	Fail to reject H_0 — past does NOT significantly predict future

Table 5. Spearman Rank Correlation Test results.

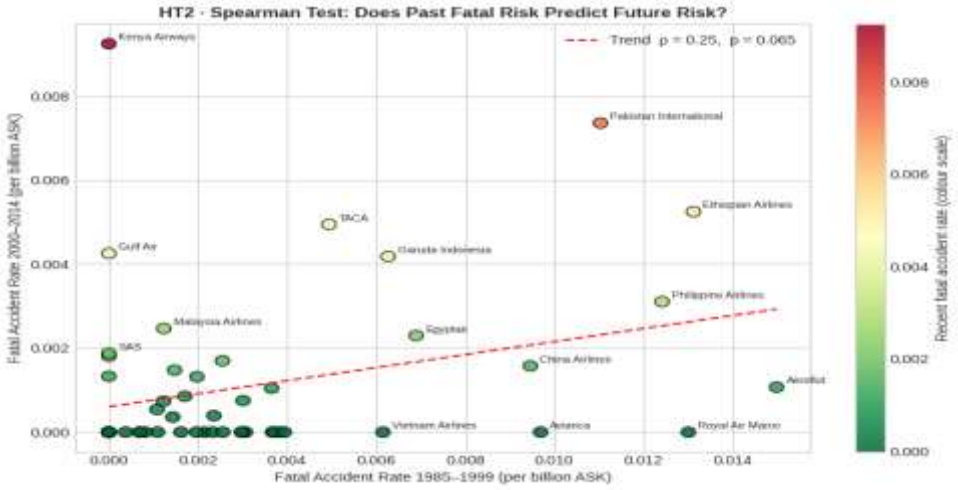


Figure 7. Scatter of fatal accident rates old vs new. Pearson r = 0.31, Spearman ρ = 0.25, p = 0.065 — weak, non-significant

Test	Statistic	p-value	Decision
Wilcoxon Signed-Rank (HT1)	W = 799.00	0.0024	Reject H_0 — safety improved significantly
Spearman Rank Correlation (HT2)	$\rho = 0.248$	0.0649	Fail to reject H_0 — past ≠ future

Table 6. Summary of formal statistical test results.

5. Composite Risk Scoring System

The composite risk score ranked Kenya Airways (0.863) and Pakistan International Airlines (0.614) as High Risk while rated Gulf Air (0.484), Ethiopian Airlines (0.381) and Malaysia Airlines (0.359) in the Moderate Risk category. The remaining 51 airlines had 91 percent operational coverage which included nine operational airlines. This includes Aer Lingus, Avianca, Finnair, LAN Airlines, and Japan Airlines. The nine airlines reported no incidents between 2000 and 2014 which resulted in a zero score. Figure 7 shows the complete risk ranking of the airlines.

Component	Metric	Weight
Incident score	Incident rate, 2000–14 (normalised)	25%
Fatal accident score	Fatal accident rate, 2000–14 (normalised)	40%
Fatality score	Fatality rate, 2000–14 (normalised)	35%
Trend penalty	+0.05 if fatality rate worsened	Added

Table 7. Composite Risk Score weighting.

Score Range	Risk Tier	Airlines (of 56)	Share
< 0.30	Low Risk	51	91%
0.30–0.59	Moderate Risk	3	5%
≥ 0.60	High Risk	2	4%

Table 8. Risk tier thresholds and distribution.

Only Kenya Airways (0.863) and Pakistan International (0.614) were High Risk. Nine airlines scored exactly 0.000. Even the highest-scoring airlines present far lower risk than ground transport.

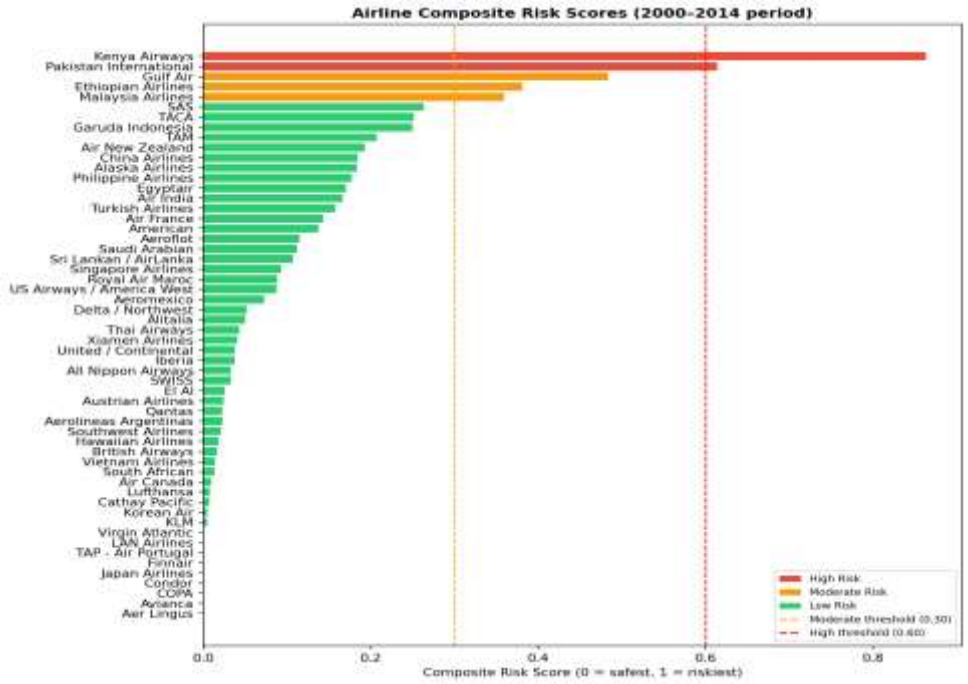


Figure 8. Composite Risk Scores for all 56 airlines. 91% are Low Risk. Dashed lines mark tier thresholds.

The pie chart below shows (figure 8) how 56 major commercial airlines distribute their operations into three risk tiers which are based on their safety performance between 2000 and 2014. The Low Risk tier includes 51 out of 56 airlines which represents 91% of the total showing that modern commercial aviation currently provides a high level of safety. The airline industry had three companies which made up 5% of the total classified as Moderate Risk while two companies which made up 4% of the total reached the High Risk classification. The distribution shows that people worry about airline safety yet their worries do not match current data which applies to almost all airlines.

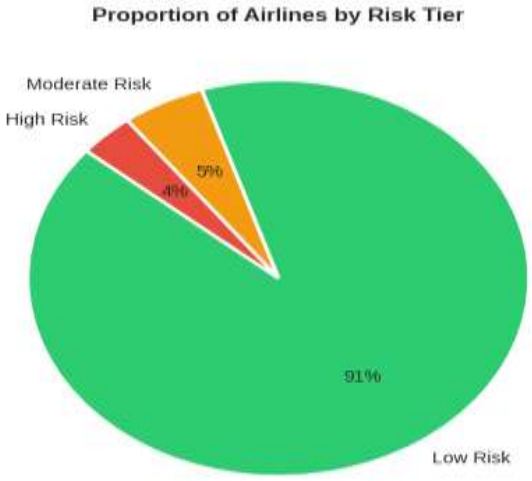


Figure 9. Proportion of airlines by composite risk tier based on 2000–2014 safety data (n = 56). Low Risk: score < 0.30; Moderate Risk: 0.30–0.59; High Risk: ≥ 0.60. The 91% Low Risk share confirms that modern commercial aviation is overwhelmingly safe.

6. Conclusions

Travellers should not avoid airlines solely based on historical crashes. The data shows safety has broadly improved across the industry and past incident counts function as weak predictors of future safety when exposure to risk has been controlled. The composite risk score establishes a logical foundation for comparison between various airlines while the dataset reveals that even the highest-scoring airlines maintain a risk level which is significantly lower than that of ground transportation.

Finding	Evidence
Safety improved significantly	Wilcoxon: $p = 0.0024$
Past is a weak predictor of future	Spearman $\rho = 0.248$, $p = 0.065$ (not significant)
91% of airlines are Low Risk	Score < 0.30
High risk driven by single events	Component score breakdown

Table 9. Summary of key findings and evidence.

References:

[1] FiveThirtyEight, "Airline Safety Dataset," 2014. [Online]. Available: https://github.com/fivethirtyeight/data/tree/master/airline-safety.

[2] F. Wilcoxon, "Individual comparisons by ranking methods," Biometrics Bulletin, vol. 1, no. 6, pp. 80–83, 1945. [Online]. Available: https://www.jstor.org/stable/3001968.

[3] C. Spearman, "The proof and measurement of association between two things," American Journal of Psychology, vol. 15, no. 1, pp. 72–101, 1904. [Online]. Available: https://www.jstor.org/stable/1412159.

[4] International Civil Aviation Organization (ICAO), "Annual Safety Report." [Online]. Available: https://www.icao.int/safety.